

CHAPTER THREE

MODELING AND CIRCUIT DEVELOPMENT

3.1 MODEL EQUATIONS OF A SEDCM

The equivalent circuit of a separately excited DC motor, shown below, is used in deriving the model equations of the motor.

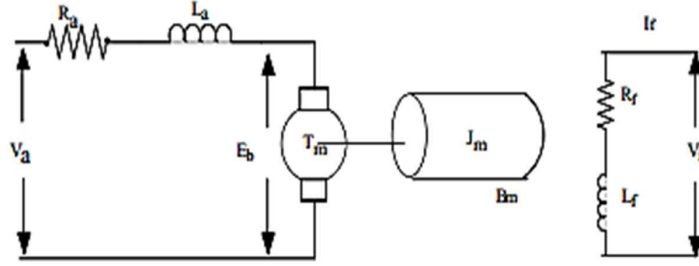


Fig. 3.1: Equivalent Circuit of a Separately Excited DC Motor

In the separately excited DC motor, the field winding is energized from a separate DC source (V_f) and the armature winding is also energized from another DC source (V_a). Armature voltage, V_a , may be variable but, a constant DC voltage source is required to energize the field winding. It is important to note that the armature and field windings are electrically isolated from each other.

3.1.1 OPEN LOOP MODEL OF A SEDCM

From figure 3.1, the field and armature voltage equations follows thus:

$$V_f = i_f R_f + L_f \frac{di_f}{dt} \quad \dots (3.1)$$

$$V_a = i_a R_a + E_a + L_a \frac{di_a}{dt} \quad \dots (3.2)$$

From Newton's 3rd Law of Motion, the torque balance equation is given as:

$$T_m = T_l + B_m \omega_m + J_m \frac{d\omega_m}{dt} \quad \dots (3.3)$$

The back emf of a DC motor is given as:

$$E_a = K_a \Phi \omega_m \quad \dots (3.4)$$

Assuming magnetic linearity, the rotor flux will be proportional to the field current, thus:

$$\Phi \approx i_f \quad \dots (3.5)$$

Hence:

$$K_a = K_m i_a = K_f i_f i_a \quad \dots (3.6)$$

Equation (3.4) becomes:

$$E_a = K_f i_f \omega_m = K_m \omega_m \quad \dots (3.7)$$

Where:

$K_m = K_f i_f$ is a constant

The torque developed by the motor is given by:

$$T_m = K_a \Phi i_a = K_m i_a \quad \dots (3.8)$$

3.1.2 TRANSFER FUNCTION OF A SEDCM

To obtain the transfer function of a separately excited DC motor, all initial conditions are set to zero and we take the Laplace transform of the motor time domain equations. The Laplace transform is independent of input excitations and shows the relationship between the input and output of the system. Hence;

From equations (3.2) and (3.7), we have:

$$V_a = i_a R_a + K_m \omega_m + L_a \frac{di_a}{dt} \quad \dots (3.9)$$

Taking Laplace transform gives:

$$V_{a(s)} = I_{a(s)} R_a + K_m \omega_{m(s)} + L_a S I_{a(s)} \quad \dots (3.10)$$

$$V_{a(s)} = K_m \omega_{m(s)} + I_{a(s)} R_a (1 + S \tau_a) \quad \dots (3.11)$$

where:

$\tau_a = \frac{L_a}{R_a}$ is the electrical time constant of the system

Making I_a subject in equation (3.11), the following is obtained:

$$I_{a(s)} = \frac{V_{a(s)} - K_m \omega_{m(s)}}{R_a(1 + S\tau_a)} \quad \dots (3.12)$$

Also, from equations (3.3) and (3.8):

$$K_m i_a = T_l + B_m \omega_m + J_m \frac{d\omega_m}{dt} \quad \dots (3.13)$$

Taking the Laplace transform:

$$K_m I_{a(s)} = T_l + B_m \omega_{m(s)} + J_m S \omega_{m(s)} \quad \dots (3.14)$$

$$K_m I_{a(s)} = T_l + B_m \omega_{m(s)} (1 + S\tau_m) \quad \dots (3.15)$$

Making $I_{a(s)}$ subject:

$$I_{a(s)} = \frac{T_l + B_m \omega_{m(s)} (1 + S\tau_m)}{K_m} \quad \dots (3.16)$$

Solving equation (3.12) and (3.16), the transfer function from the input voltage $V_{a(s)}$ to the output angular velocity $\omega_{m(s)}$ is obtained as follows:

$$\frac{\omega_{m(s)}}{V_{a(s)}} = \frac{K}{(R_a + SL_a)(SJ_m + B_m) + K^2} \quad \dots (3.17)$$

where:

$$K = K_e = K_t$$

The block diagram representation of the above system is shown below:

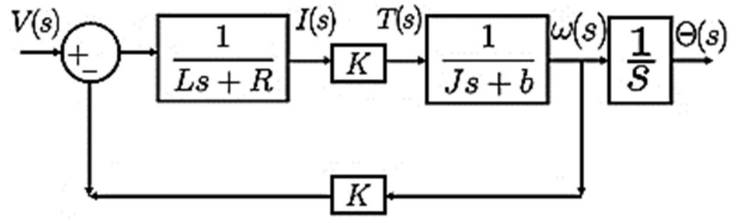


Fig. 3.2: Block Diagram of representation of the Transfer Function [16]

3.1.3 CLOSED LOOP MODEL OF A SEDCM

In the closed loop model, a PID controller is inserted into the machine for the speed control. The diagram of the system is as shown below:

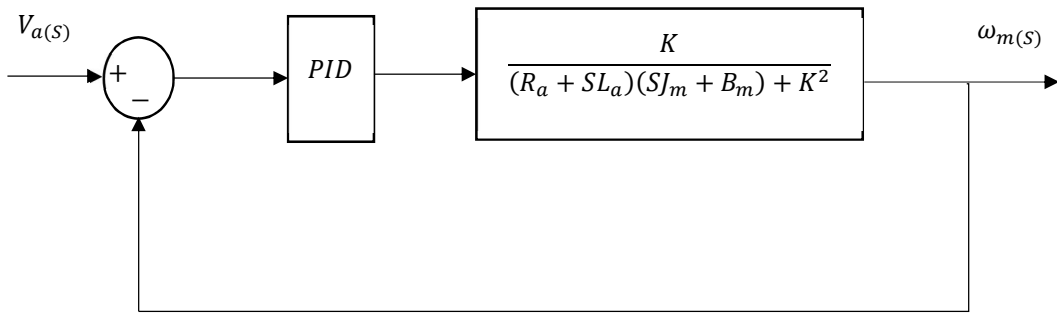


Fig. 3.3: Block Diagram of the Closed Loop Model

For the PID, the relationship between the input and the output is given as:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt} \quad \dots (3.18)$$

Taking the Laplace transform gives:

$$U(S) = K_p E(S) + K_i \frac{E(S)}{s} + K_d s E(S) \quad \dots (3.19)$$

In terms of the time constant, τ , equation (3.19) becomes:

$$U(s) = K_p \left[1 + \frac{1}{\tau_i} + \tau_d \right] E(s) \quad \dots (3.20)$$

Where τ_i and τ_d are the differential and integral time constants given as:

$$\tau_i = \frac{K_p}{K_i} \quad \dots (3.21)$$

$$\tau_d = \frac{K_d}{K_p} \quad \dots (3.22)$$

The overall transfer function of the closed loop system is obtained as follows:

$$T = \frac{G_c G_p}{1 + G_c G_p} \quad \dots (3.23)$$

where G_c is the controller transfer function, which in this work is the *PID* controller, and G_p is the process transfer function which is represented by the DC motor transfer function. Putting equations (3.17) and (3.19) into (3.23), the transfer function of the motor with *PID* controller is as given below:

$$G_{(s)} = \frac{(K_d S^2 + K_p S + K_i)K}{L_a J_m S^3 + (R_a J_m + B_m J_m + K_d)S^2 + (R_a B_m + K_p + K) + K^2} \quad \dots (3.24)$$

The overall motor model with the *PID* controller is as shown in fig. 3.4 below. *PID* controller brings the best of the entire available controller. The proportional and integral controller decreases the rise time & eliminate the steady state error while the derivative controller decreases the overshoot and settling time which is increased by the proportional and integral gain.

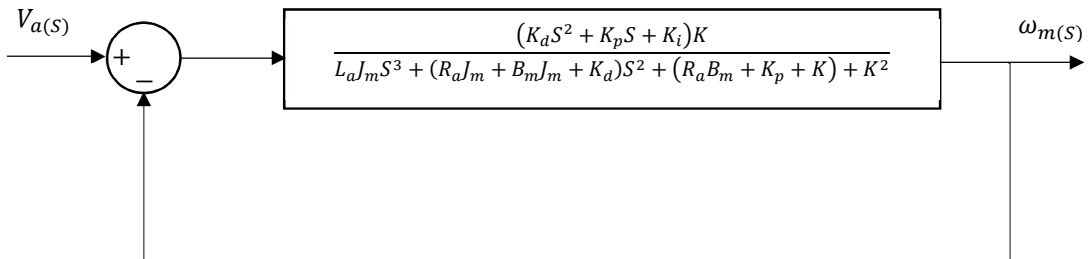


Fig. 3.4: SEDC Motor Model with PID Controller

3.2 SIMULATION EQUATIONS

The equations for modelling of a separately excited DC motor are obtained from the primary equations above. These equations are used to obtain the parameters of the motor which are time-variant (varies with time). Such parameters include the motor speed, field current, armature current and the motor torque, which varies as the load in the motor varies.

From equation (3.1), the field current is obtained by making the derivative subject:

$$\frac{dI_f}{dt} = \left(\frac{1}{L_f}\right)V_f - \left(\frac{R_f}{L_f}\right)I_f \quad \dots (3.25)$$

Also, from equation (3.2), the armature current is obtained by making the derivative subject, and substituting equation (3.6) into (3.2):

$$\frac{dI_a}{dt} = \left(\frac{1}{L_a}\right)V_a - \left(\frac{R_a}{L_a}\right)I_a - \left(\frac{K_a}{L_a}\right)(\omega_m \times I_f) \quad \dots (3.26)$$

Making the derivative of speed the subject in equation (3.3) gives:

$$\frac{d\omega_m}{dt} = \frac{1}{J_m}(T_m - T_l - B_m\omega_m) \quad \dots (3.27)$$

Substituting $K_m = K_a I_f$ in equation (3.8), the motor torque is obtained as follows:

$$T_m = K_a I_f I_a \quad \dots (3.28)$$

Equations (3.25), (3.26), (3.27), and (3.28) are the basic modelling equations for a separately excited DC motor, and the parameters are defined as below:

Φ = flux

T_m = motor torque

I_f = field current

I_a = armature current

K_a = motor constant

J_m = moment of inertia of the motor

T_l = load torque

B_m = viscous coefficient

ω_m = motor speed

V_a = armature excitation voltage

L_a = armature inductance

L_f = field inductance

E_a = back emf

V_f = field excitation voltage

3.3 MOTOR PARAMETERS

The parameters of the separately excited DC motor used in the simulation is given in table 3.1 below. The motor transfer function is obtained by substituting the values into equation (3.17).

Table 3.1: Simulation values of the DC motor

Parameter	Value
R_a	2Ω
L_a	16.2mH
R_f	180Ω
L_f	71.47H
J_m	0.117kg.m^2

B_m	0.001N.m. s
T_l	0-80N.m
K_b	0.699Nm/A ²
V_a	220V _{DC}
V_f	300V _{DC}

From equation (3.17), substituting the values of R_a , L_a , J_m , B_m , and $K_b = K$, the motor transfer function becomes:

$$\frac{\omega_m(S)}{V_a(S)} = \frac{0.6990}{0.001895S^2 + 0.23402S + 0.490601} \quad \dots (3.29)$$

3.4 MODEL SIMULATION

The individual block model of the basic equations, i.e. (3.25), (3.26), (3.27), and (3.28), in Simulink is shown below.

For the motor armature current: $\frac{dI_a}{dt} = \left(\frac{1}{L_a}\right)V_a - \left(\frac{R_a}{L_a}\right)I_a - \left(\frac{K_a}{L_a}\right)(\omega_m \times I_f)$

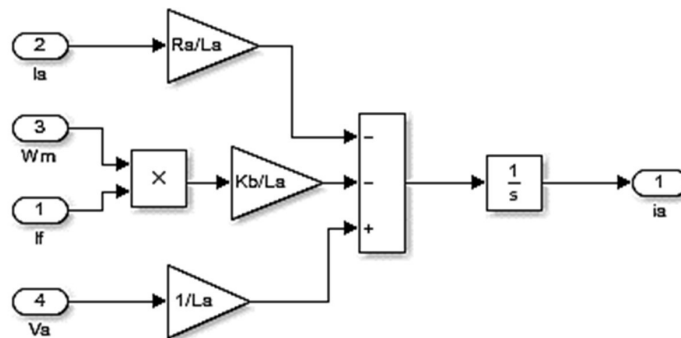


Fig. 3.5: Armature Current Model

For the motor field current: $\frac{dI_f}{dt} = \left(\frac{1}{L_f}\right)V_f - \left(\frac{R_f}{L_f}\right)I_f$

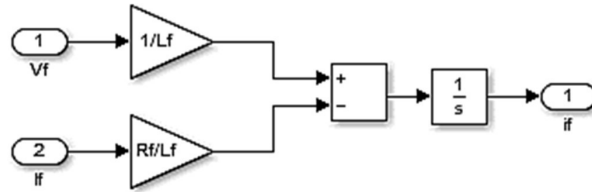


Fig. 3.6: Field Current Model

For motor speed: $\frac{d\omega_m}{dt} = \frac{1}{J_m}(T_m - T_l - B_m\omega_m)$

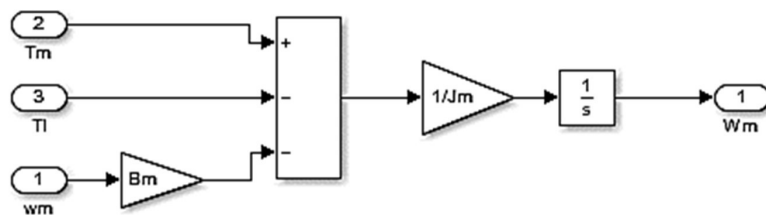


Fig. 3.7: Motor Speed Model

For motor torque: $T_m = K_a I_f I_a$

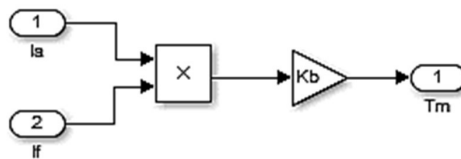


Fig. 3.8: Motor Torque Model

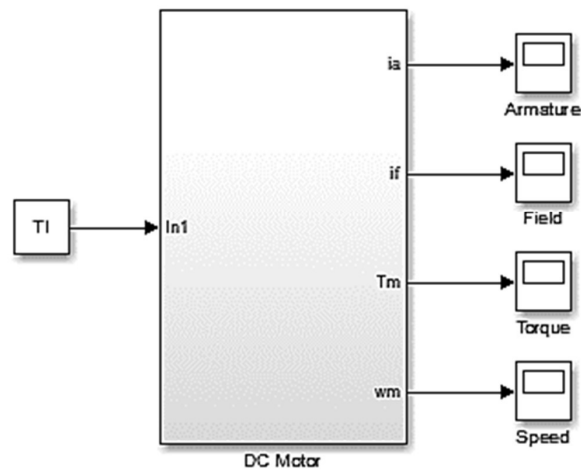


Fig. 3.9: DC Motor Model

For the closed loop model, the *PID* controller is added in the motor through the armature voltage terminal. The model is as shown below:

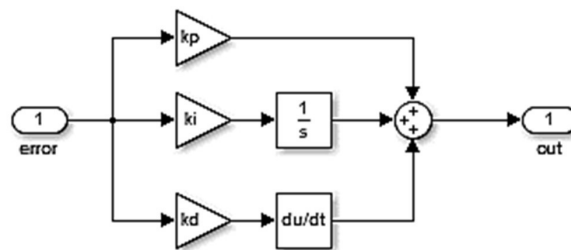


Fig. 3.10: *PID* controller model

In applying the controller to the DC motor, the motor transfer function given in equation (3.29) is used and by applying a unit step response at the input, the motor operation can be observed. The figure below shows the open loop transfer function motor simulation.

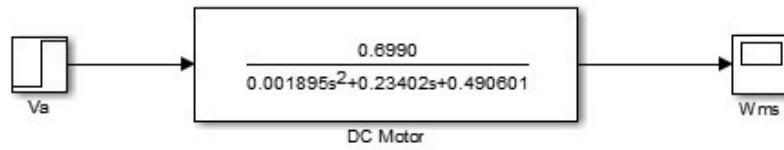


Fig. 3.11: Open Loop Transfer Function Simulation

The *PID* controller in fig 3.10 is used in the motor and a feedback incorporated for proper tracking of the reference point.

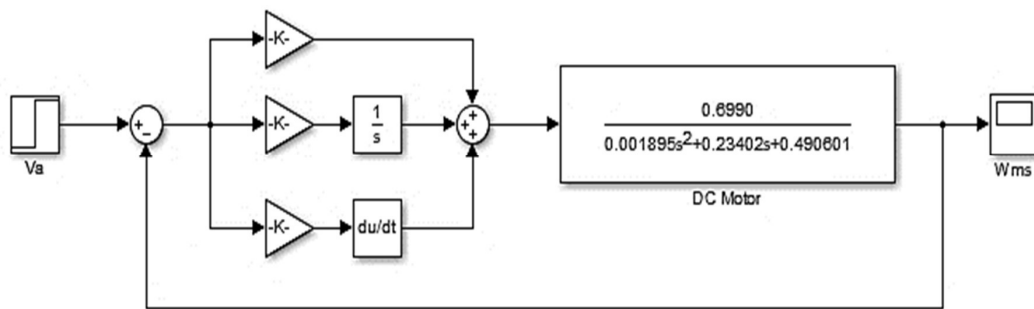


Fig. 3.12: Closed Loop Transfer Function Model

3.5 PID PARAMETERS

After running the MATLAB code for the machine in the MATLAB command window and using the *PID* tuner, taking a fast response time and a robust system since we want the motor to settle fast to its steady state, the following values obtained are as shown in table 3.2 below.

Table 3.2: *PID* Parameters

Rise Time	0.0061 seconds
Settling Time	0.0238 seconds
Overshoot	4.62%
K_p	131.24
K_i	633.63
K_d	0.71

With the above values obtained for the *PID* parameters, the motor transfer function with the controller, equation (3.24), is given as:

$$G(S) = \frac{0.4963 s^2 + 91.74 s + 442.9}{0.0019 s^3 + 0.7303 s^2 + 92.23 s + 442.9} \quad \dots (3.30)$$

The transfer function of equation (3.30) was also simulated to compare with the result obtained from fig. 3.12.

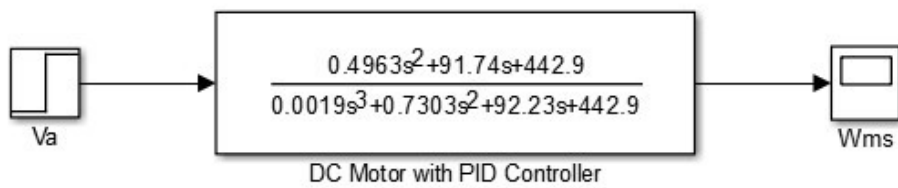


Fig.3.13: Over all Transfer function

Disturbances were added in the system in the form of loads and the motor response was observed. These disturbances were added in time intervals of 1, 2.3 and 3.8 seconds with values of 0.5, -0.7 and 0.9 respectively for a simulation time

of 5 seconds. The figure below shows the point the disturbances were added to the system.

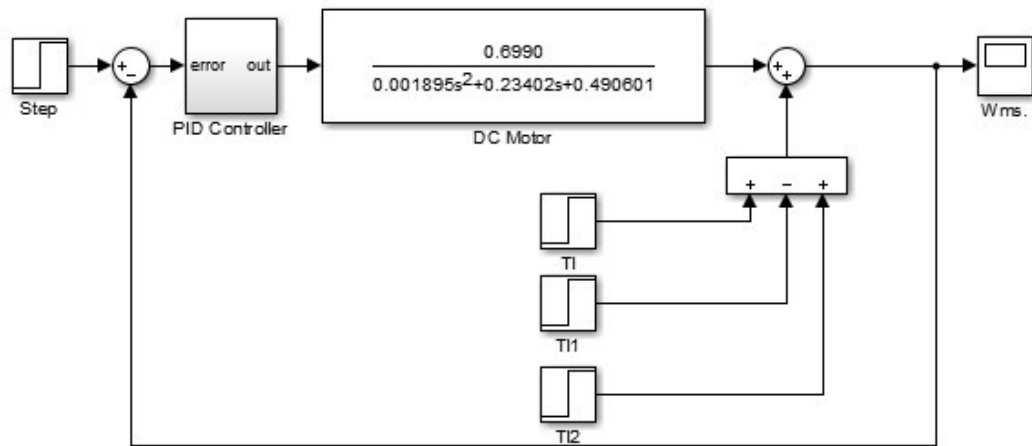


Fig. 3.14: Adding disturbances into the system

CHAPTER FOUR

RESULTS AND ANALYSIS

The motor simulation results are as shown below. These simulations were carried out at variable loads between 0N.m to 80N.m. This was done in order to observe the motor characteristics at various load conditions. The simulation was done for 10 seconds and the loads were introduced at 3 seconds, 5 seconds and 7 seconds.

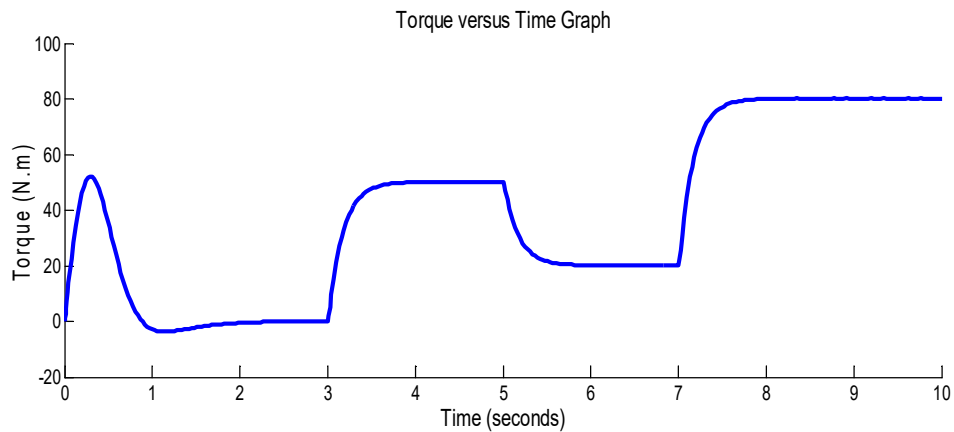


Fig. 4.1: Variable Input Torque

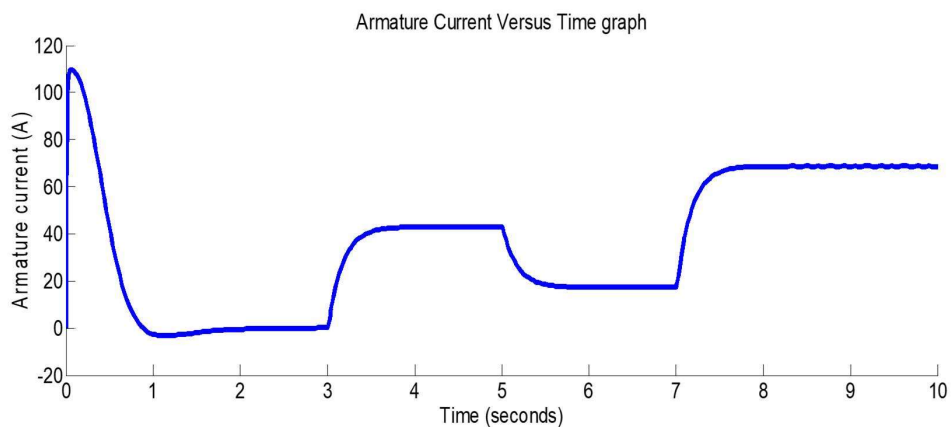


Fig. 4.2: Armature Current

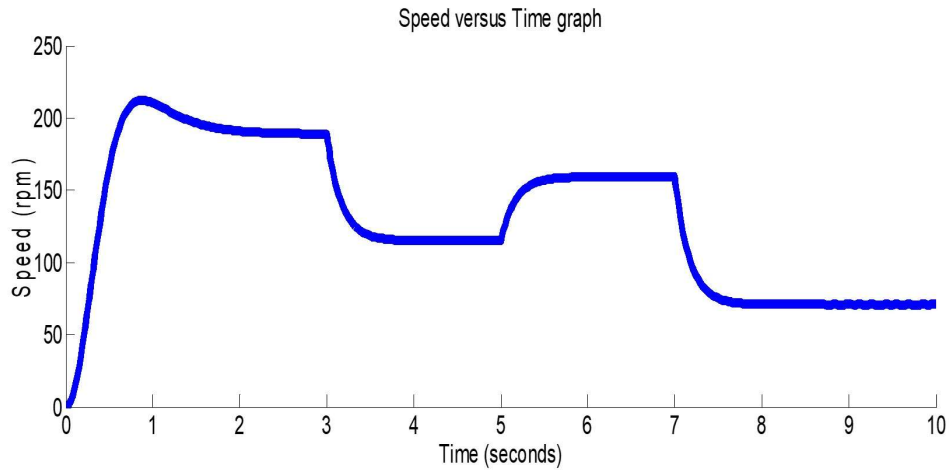


Fig. 4.3: Motor Speed

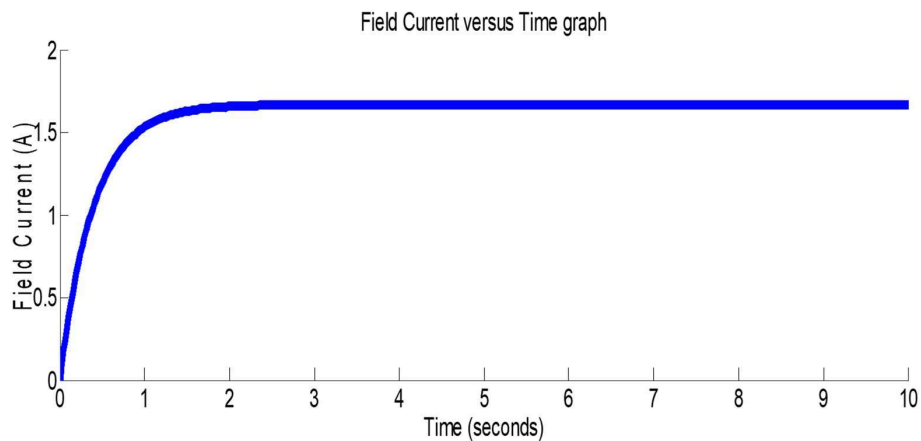


Fig. 4.4: Field Current

In the simulation, a load of 50N.m was added at 3 seconds, then followed by a reduction in the load by 30N.m and another 60N.m increase at the 5th and 7th seconds respectively. This load increase can be seen in the torque curve in fig. 4.8 above. As observed, the motor's armature current increases with increase in the load, thus having a similar curve. Only the motor speed does the opposite. As the load is increased, the armature current increases also but the speed of the motor reduces. This is as a result of no controller action. The motor draws very high current but the speed reduces, thus reducing the efficiency of the motor. This

is what necessitated the use of a PID controller in order to keep the motor speed constant as the load is increased. Fig. 4.11 is the motor field current and as can be observed from the graph, it is not affected by the load changes. This is because it has a separate voltage source from that of the armature, hence the name separately excited DC motor.

4.1 OPEN LOOP MODEL

The motor transfer function was also simulated with a unit step input to its behaviour before applying the controller. The result below was obtained.

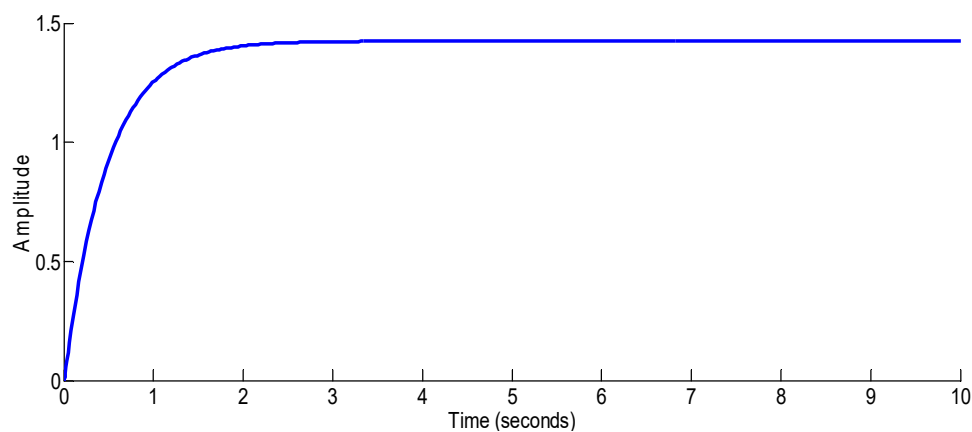


Fig. 4.5: Open Loop Response

As can be seen from the plot of the motor's unit response, there is a very high difference between the step response and motor output. This is because there is no feedback loop and a controller to track the reference and give desired output.

4.2 CLOSED LOOP MODEL

The closed loop model given in fig. 3.12 and 3.13 were both simulated to observe the motor's tracking ability with the PID controller in place. The following results were obtained from figs. 3.12 and 3.13 respectively.

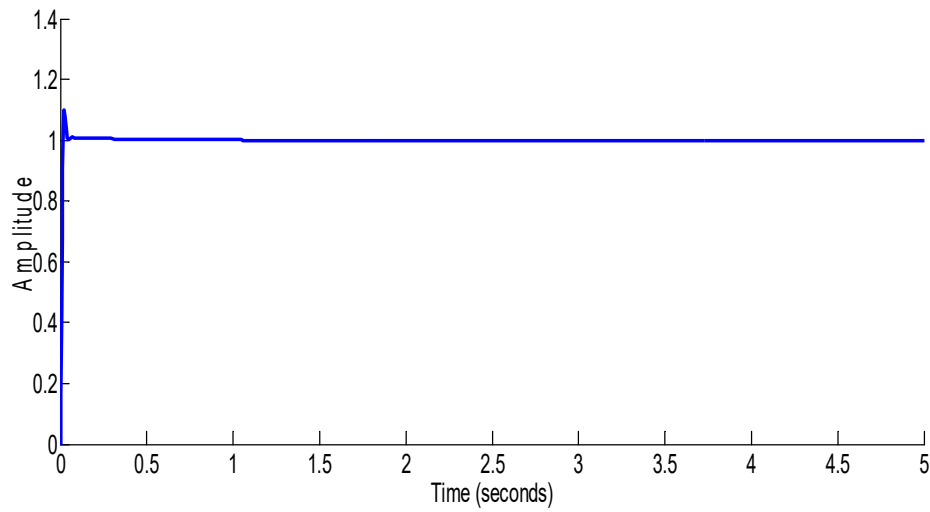


Fig. 4.6: Closed Loop Model with Separate PID (fig. 3.12)

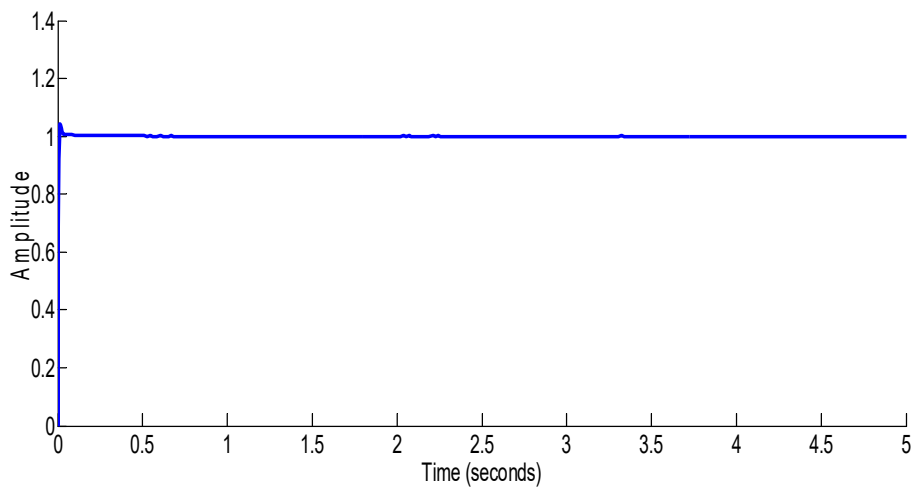


Fig. 4.7: Closed Loop Model with Embedded PID (fig. 3.13)

As observed from the motor response with the controller, the motor has a fast settling time and tracks the reference properly.

When disturbances were introduced into the system with the PID controller, the result below was obtained:

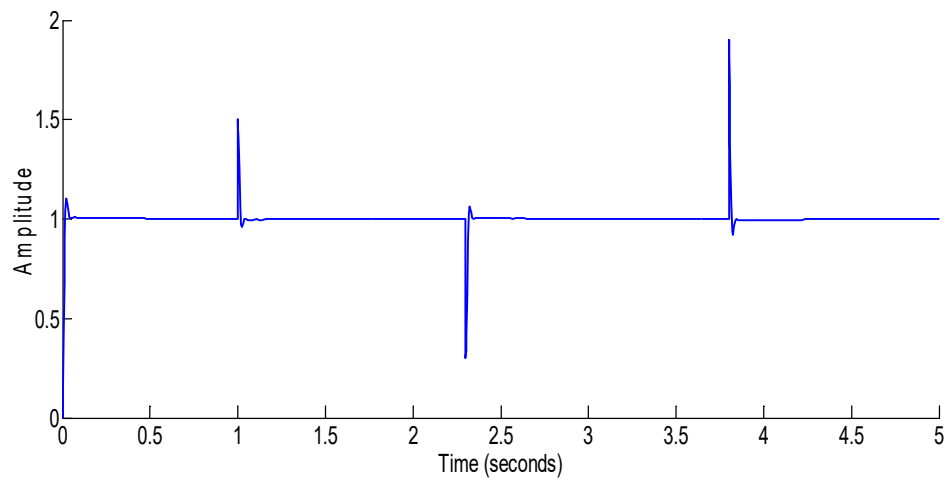


Fig. 4.8: Motor response to disturbances in the system

The PID controller always regulates the motor output and tracks the reference in order to maintain the required output. The PID, is thus good for proper system regulation in industrial applications.

CHAPTER FIVE

5.1 CONCLUSION

In the work, the PID control of a separately excited DC motor have been successfully studied. The results obtained shows a better performance of the motor than when in open loop. When loaded, the motor characteristic changes but the PID controller returns the motor back to its normal operation. The rise time, overshoot and steady state are also minimized as compared to the *PI* and *PD* controllers.

5.2 RECOMMENDATION

Though being efficient and widely used, the PID controller can still be modified by adding a fuzzy logic controller to it or using fuzzy logic totally for the speed control of a separately excited DC motor.